Sahaj Bhadja

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Summary

Frontend-leaning Software Engineer with 2+ years building **React** and **TypeScript** interfaces for data-heavy operations and scheduling workflows, fast dashboards over messy real-world data, and **Python** backends, with an eye for clear, mistake-proof **UI**.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Cut dashboard latency from 1.2s to 350ms for 500+ users by redesigning multi-tenant schemas and indexed query access patterns behind the **data-heavy dashboards** of an AI education platform.
- Built **Python** integration pipelines that **validate and reconcile** messy records across PostgreSQL, SQL Server, and cloud storage, sustaining 99%+ reliability across 5K+ monthly jobs with incremental loads and error handling.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Designed the **React** frontend flow for question upload, rubric generation, grading, and grading history, wiring dashboard views to Flask REST endpoints used by 150+ instructors handling 10K+ daily requests.
- Kept the UI **responsive** during large uploads by moving grading to multithreaded batch processing, and surfaced scores, feedback, and AI rationale clearly, cutting instructor feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Reduced payloads for low-bandwidth clients with pagination, field filtering, and caching of frequently requested data, improving **response times** 30% across large-scale API integrations in Agile sprints with Microsoft stakeholders.
- Automated 20+ unit and integration test suites with PyTest and Mockito across Windows, Linux, and macOS, and **refactored** modules with Factory and Singleton patterns to cut technical debt 20%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built a **React and TypeScript** resource-allocation and operations platform with real-time WebSocket updates for assignments, conflicts, billing, and leads, syncing 2M+ records across locations with 33% fewer sync conflicts.
- Studied feature usage, load times, and drop-off points in Google Analytics to fix **confusing interface flows** and slow pages, helping cut resource wastage 42% and lift lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built a React and TypeScript web dashboard with **time-slot booking**, real-time Firebase status updates, and Flask APIs under 200ms, using locking to prevent double-booked slots across 50K+ bookings.
- Designed system flows in **Figma**, reducing design errors 25%, and presented the end-to-end architecture to the client while improving cross-platform compatibility 15% across Android, iOS, and web.

Skills

Languages: TypeScript, JavaScript, Python, HTML, CSS, SQL, Java
Frontend: React, Next.js, TailwindCSS, Angular.js, WebSockets, Responsive Design, EJ-2 Gantt, Figma
Backend & Data: Node.js, Express.js, FastAPI, Flask, REST APIs, PostgreSQL, SQL Server, MongoDB, Firebase
Product & UX: UI/UX Design, Data Visualization, Tableau, Performance Optimization, Google Analytics
Tools & Practices: Git, GitHub Actions, Docker, AWS, Vercel, Jest, Selenium, PyTest, Postman, CI/CD, Agile

Projects

Production Schedule Gantt Explorer | *React, TypeScript, Next.js, Tailwind, Canvas, D3*

- Designed a **canvas-rendered** Gantt view in React and TypeScript for thousands of operations across machines, with capacity heatmaps, due-date risk flags, and drag-to-reschedule that validates **constraints** before saving.

Plan Scenario Comparison Workspace | *Next.js, TypeScript, Tailwind, SVG, Python, FastAPI*

- Designed a Next.js workspace for branching what-if **scenario trees**, comparing late orders and utilization side by side in SVG charts, with explicit loading, empty, stale, and **malformed-data states** for messy ERP imports.

CPU-Scheduling Simulator Live | *React.js, JavaScript, C++, EJ-2, OS Scheduling*

- Built an interactive **scheduling** simulator with React controls and an EJ-2 **Gantt chart** visualizing FCFS, SJF, SRT, Round Robin, and MLQ runs, with live metric updates that cut debugging time 20%.